Questions Exercise 4

Employees

Employee_id	First Name	Last Name	Department	Salary	City	Age
1	John	Doe	IT	70000	New York	35
2	Alice	Smith	HR	60000	Los Angeles	40
3	Bob	Johnson	Finance	75000	Toronto	45
4	David	Brown	IT	72000	London	30
5	Emma	Wilson	Sales	65000	Sydney	28
6	Michael	Clark	Finance	80000	New York	50

Projects

Project ID	Emp_id	Project name	Budget	Start date	Status
101	1	AI Development	100000	2024-01-16	Completed
102	2	Employee Training	50000	2024-02-15	Ongoing
103	1	Cybersec Audits	75000	2024-03-05	Pending
104	3	Financial Analysis	90000	2024-04-12	Ongoing
105	5	Market Expansion	65000	2024-05-20	Completed
106	6	Risk Management	80000	2024-06-10	Pending

1. Retrieve all employees and their assigned projects using an Inner Join.
   - Return: EmployeeID, First Name, Last Name, Department, Salary, ProjectID, Project Name, Budget, Status

```
SELECT   employee ID, Last Name, Department, Salary, Project ID,
         First Name, Project Name, Budget, Status
FROM     Employees AS A
INNER JOIN  Projects AS B
ON       A.employee id = B.employee ID
```

Employ-ID	First Name	Last Name	Department	Salary	Project ID	Project Name	Budget	Status
1	John	Doe	IT	70k	101	AI Develop	100k	completed
2	Alice	Smith	HR	60k	102	Employee Training	50k	Ongoing
3	Bob	Johnson	Finance	75k	104	Financial Analysis	90k	Ongoing
5	Emma	Wilson	Sales	65k	105	Market Expansion	65k	Completed
6	Michael	Clark	Finance	80k	106	Risk management	80k	Pending
1	John	Doe	IT	70k	103	Cyber Sec Aud	75k	Pending

2. Retrieve all employees and their assigned projects, including employees who have no projects using a LEFT JOIN

- Return: Employee ID, First Name, Last Name, Department, Salary, Project D Project Name, Budget Status.

```
SELECT   employee ID, First Name, Last Name, Department, Salary, Project ID
         Project Name, Budget, status

FROM     Employees AS A
LEFT JOINS   Projects AS B
ON       A. employee ID = B. employee ID
```

~~Employee~~ ID
~~1~~

number 2 & 4 Tables are the same

Emp-id	First Name	Last Name	Department	Salary	Project ID	Project Name	Budget	Status
1					101	AI Development	100k	Completed
2	"	"	"	"		"		"
3								
4	David	Brown	IT	72000	NULL	NULL	NULL	NULL
5								
6	"	"	"	"		"		"
1					103	CyberSecAudits	75k	Pending

4. Retrieve all employees and projects, including those without a match in either table using a FULL OUTER JOIN

Miro, GitHub, Canva ~ Project Plan Google Sheet, Vessel